

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Systems for Data Science, Performance Engineering, Optimization in CS, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, Java, JavaScript, TypeScript, SQL, Bash

ML & Deep Learning

PyTorch, JAX, computer vision (CNNs, ResNet), normalizing flows, diffusion models, reverse-mode autodiff, NumPy, pandas, scikit-learn, HuggingFace Transformers

Data & Pipelines

Apache Spark / PySpark, Spark UI, FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB, SQLite

Retrieval & LLM Systems

RAG, cross-encoder reranking, MMR, LLM agents, OpenAI / Groq APIs, Ollama

Cloud & DevOps

AWS, Docker, Kubernetes, GCP, Azure, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.
- Served as primary technical point of contact for the eTriage backend, partnering with clinical and product teams across releases and client integrations, debugging production issues, and running knowledge transfer for new engineers.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

Generative Models & Deep Learning - Flows, Diffusion, CIFAR-10 Vision

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented **normalizing flows with coupling layers** and a **DDPM diffusion model** from scratch in JAX, deriving the ELBO and closed-form Gaussian-KL objectives via variational inference, hands-on with the generative-AI methods now driving SOTA systems.
- Trained and benchmarked **5 neural architectures** (perceptron, deep MLP, ReLU MLP, VGG-style CNN, ResNet) on **CIFAR-10 image classification** across 3 optimizers with learning-rate tuning, and built a reverse-mode **automatic differentiation** engine from scratch in NumPy, validated against JAX to machine precision.

Spark ETL Optimization - Large-Scale Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI | 1.4B-row NYC TLC dataset

- Cut end-to-end ETL runtime **25%** (17–19 min to 14.3 min) on a **1.4B-row** dataset by replacing shuffle joins with broadcast joins (minutes to 0.05s), enabling AQE and skew-join handling, and tuning shuffle partitions, schema-aware reads, and year/month partitioning with Snappy compression.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition pruning, column pruning, and filter pushdown; used **Spark UI** stage timings to localize shuffle and skew bottlenecks and document a repeatable optimization playbook.
- Added a 6-rule **data-quality filter** (retaining **95.26%** of records) and a parallel data downloader, keeping the dataset clean and the pipeline reproducible across naive and tuned benchmark runs.

AgenticSearch - Automation Pipeline with Verification & Provenance

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **automation pipeline** that plans typed retrieval, reranks results, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns **cell-level provenance** for every value, quality-checking AI output instead of trusting a single opaque call.
- Hardened the service with **150+ automated tests**, provider routing/fallback, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.